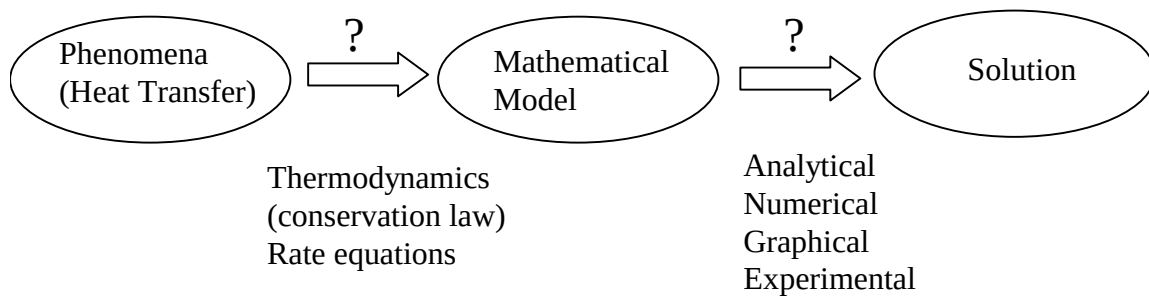


2. Basic methodology

(1) Introduction

Suppose you have a problem involving heat transfer. For example, you may want to estimate the temperature increase in a thermal system. To analyze the given problem, the first step is to translate the problem into mathematical terms. In other words, you should develop mathematical expressions that mimic the phenomena. The process is called modeling. In HT, there are two main tools for modeling. The first is the rate (constitutive) equations (Fourier's law, Newton's law of cooling, Stefan-Boltzmann equation). Once you identify the HT mechanism(s), i.e., the corresponding rate equation, the second step is considering the conservation principle (conservation of mass, momentum and energy). The conservation laws together with the rate equations will lead to a mathematical expression, i.e., a model. The model can be a differential/integral/algebraic equation(s). Now, you can work on the model (governing equation) to get the solution using analytical, numerical, and/or experimental methods.



(2) Conservation of energy (Energy balance)

- The law of conservation of energy tells you that energy does not disappear. For a specific system, the overall energy balance should be maintained considering the energy that the system gains or loses.
- 1st law of thermodynamics: $Q = \Delta E + W$ or $\dot{Q} = \Delta \dot{E} + \dot{W}$
Net heat transfer (rate) = Store energy (rate) - Work done by the system (power generated)
- In HT, there's no work involved. Therefore, net heat transfer or heat generation = Stored energy. $Q = E$, i.e., $\Delta E_{st} = E_{in} - E_{out} + E_g$ in HT terms.

Differentiating with respect to time, $\boxed{\dot{E}_{st} = \dot{E}_{in} - \dot{E}_{out} + \dot{E}_g}$.

Potential or kinetic energy is often negligible in comparison with thermal (internal) energy, in which case, $\dot{E}_{st} = \frac{dU}{dt} = \dot{E}_{in} - \dot{E}_{out} + \dot{E}_g$.

$\Delta U = mc\Delta T$ or $\frac{dU}{dt} = mc \frac{dT}{dt} = \rho c V \frac{dT}{dt}$ without phase change. If phase change occurs,

$\Delta U = mc\Delta T + mh_l$ (c : specific heat, h_l : latent heat of vaporization, melting, ...)

$\dot{E}_{in}$, and $\dot{E}_{out}$ are calculated by using proper rate equations, e.g., $\dot{E}_{in} = -kA \frac{dT}{dx} \Big|_{in}$, and

$\dot{E}_{out} = hA(T - T_\infty)$.